MILITARY SPECIFICATION

BLOWER, GASOLINE-ENGINE-DRIVEN, TYPE A-1

1. SCOPE

1.1 This specification covers one type of gasoline-engine-driven blower, designated Type A-1.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-P-664	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated List
MIL-L-2104	Lubricating Oil, Internal Combustion Engine (Heavy Duty)
MIL-G-3056	Gasoline, Automotive, Combat
MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion On
MIL-H-4607	Heater, Engine And Shelter, Ground, Type H-1
MIL-C-5541	Chemical Films And Chemical Film Materials For Aluminum And Aluminum Alloys
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-L-6082	Lubricating Oil; Aircraft Reciprocating Engine (Piston)
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For
MIL-E-7729	Enamel, Gloss, For Aircraft Application
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

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MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-E-11275	Engines, Gasoline, Industrial Type, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking And Waterproofing; With Appropriate Test Methods
MS33586	Metals, Definition Of Dissimilar

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

166	Colors; List Of Standard Aircraft Glossy

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

PTC 9-1954	Displacement Compressors, Vacuum Pumps and Blowers
PTC 19.5:4	Supplement on Instruments and Apparatus; Part 5, Chapter 4, Flow Measurement by Means of Standardized or Orifice Plates

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

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* 3.2 Components. The blower shall consist of the following major components:

ITEM NO.	QUANTITY	ITEM OR SYSTEM NAME
1	1	Compressor
2	1	Intake Filter
3	1	Discharge Header
4	1	Gasoline Engine
5	2	Carrying Handles
6	1	Metal Hood

* 3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exceptions in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Dissimilar metals. Insofar as practicable, dissimilar metals in contact with each other shall be avoided. However, to provide similar or suitable abutting surfaces, metal plating or spraying of dissimilar base metals shall be permitted. The use of dissimilar base metals separated by suitable insulating material shall also be permitted. Dissimilar metals are defined in MS33586.

3.4.2 Protective treatment. When materials are used in the construction of the blower that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

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3.5 Design and construction. The blower shall be a compact, self-contained unit designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, or other conditions incident to shipping, storage, installation, and service.

3.5.1 Adjustments and repairs. The blower shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases without the use of special tools. Arrangement and configuration shall be such that all components can be easily removed and replaced.

3.5.2 Mounting. The blower shall be mounted on a metal base plate. Shock mounting shall be used to eliminate creeping and vibration. Mounting bolts shall be accessible and securely fastened with lock washers or shall be lock wired.

3.5.2.1 Base plate. The base plate shall be of the skid type so that the blower can be easily pulled over loosely packed sand snow.

3.5.2.1.1 Provisions shall be made on the base plate for the inclusion of tiedown rings, D-type carrying handles, and attachment of a metal hood. The minimum inside radius of the tiedown provisions shall be 1-3/4 inches. The tiedown and lifting provisions may be combined.

3.5.3 Acceleration forces. The blower shall be constructed to withstand continuous forces equal to an acceleration of 3 gravity (G) along each of the three major axes and shall operate satisfactorily after subjection to this force.

3.5.3.1 The blower shall be so constructed that all components will be entirely restrained from causing any external damage in the event that acceleration forces of 8G in either direction along the horizontal axis or 4-1/2G downward are encountered. Tiedown provisions shall not fail under these forces; however, the blower need not be operational after such force.

3.5.4 Life. The blower shall be designed for an operating life of 500 hours between major overhauls and shall be capable of operating for 300 hours without failure of any major part (see 4.5.7).

3.6 Performance. The blower shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -65° to +130° Fahrenheit (F) and ambient storage temperatures ranging from -80° to +160°F.

b. Pressure altitudes ranging from sea level to 6,000 feet.

c. Rainfall as encountered in any locale.

d. Sand and dust as encountered in desert operation.

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e. Exposure to salt-laden atmosphere.

f. Fungus growth as encountered in tropical climates.

3.6.1 Tilted position. The blower shall function satisfactorily when its normal horizontal plane is at a 15 percent (8-1/2°) angle in any direction from the true horizontal. This particularly applies to carburetion and lubrication.

3.6.2 Low temperature starting. The blower shall be capable of starting and operating satisfactorily at 0°F without oil dilution or external heating aids. The blower shall be capable of starting and operating continuously at temperatures down to and including -65°F with the use of external heat.

3.7 Detail of components.

3.7.1 Compressor. The compressor shall be of a single-stage, air-cooled type.

3.7.2 Capacity (actual delivery). The compressor shall be capable of supplying a minimum of 50 cubic feet per minute (cfm) of free air at a manifold pressure of 5 pounds per square inch gage (psig). The free air delivery shall be based on compressor inlet conditions as specified in ASME power test code 9-1954.

3.7.3 Discharge pressure. The compressor shall be designed to maintain a continuous discharge pressure of 5 psig.

3.7.4 The compressor shall be rigidly connected to the gasoline-engine equipment mounting pads and shall be provided with a relief valve, discharge header, and necessary accessories.

3.7.5 The compressor drive shall be connected to the engine either by means of a flexible coupling or through a gear case. If a gear case is used, it shall be an integral part of the compressor and shield.

3.7.6 The compressor shall have an oil-filler hole, located in an easily accessible position. No outside visible oil sight need be provided.

3.7.7 Lubrication. The compressor shall be provided with a lubrication system of such oil capacity and consumption that it will not be necessary to add oil more often than every 25 hours. Oil changes shall not be necessary more often than every 150 hours.

3.7.8 Oil drainplug. The oil drainplug shall be of the magnetic type and shall be at least 1/4-inch pipe size. It shall be located in an easily accessible position so that no tipping of the unit will be necessary for draining.

3.7.9 Oil-level hole. The oil-level hole shall be fitted with a removable pipe plug to permit checking the oil level. The oil-level hole shall not be in direct vertical line with the oil-filler hole, unless precautions are taken to insure that oil, when poured into the filler hole, will not run directly out of the level hole, thus giving a false indication.

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3.8 Intake filter. A dry-type intake filter shall be provided and shall be securely fastened in place. It shall be capable of removing all particles which may cause damage or excessive wear to the blower. The filter shall be easily removable for cleaning and shall be of such capacity that cleaning will not be required for operating periods of less than 50 hours under normal conditions.

3.9 Discharge header. A readily accessible discharge header shall be provided. The header shall be compact and shall be designed to permit operation of the discharge valves. Four valves shall be provided, each having 1-inch National Taper Pipe (Thread) (NPT) internal pipe threads. The type of valves to be used shall not permit leakage, shall operate freely under any of the environmental conditions specified herein, and shall fit within the envelope dimensions of the unit in both the open and closed positions. All header mounting bolts shall be locked wired in place.

3.9.1 Relief valve. An adjustable low-pressure, pop-off type relief valve shall be provided on the discharge header and shall be set at a cracking pressure of 4 pounds per square inch (psi). Adjustment range shall be approximately ±50 percent of the 4 psi setting with zero leakage virtually to the cracking pressure. Reset pressure, dead tight, shall normally be within 15 percent of cracking pressure. The relief valve shall be capable of passing the total amount of air supplied by the compressor at an engine speed of 3,600 revolutions per minute (rpm) without creating a pressure of more than 4-1/2 psi when set at the 4 psi cracking pressure.

3.9.2 Check valve. A check valve shall be provided between the header and compressor to prevent deflation of the lifting bags in the event of engine or compressor failure.

3.10 Gasoline engine. The compressor shall be driven by a lightweight, air-cooled, gasoline engine conforming to MIL-E-11275, with the exceptions specified herein. When the compressor is delivering air as specified in 3.7.1, the engine, at sea level, shall not be loaded in excess of 75 percent of full throttle horsepower at the same speed and inlet carburetor air temperature. The engine shall be provided with a governor, carburetor, magneto, fuel tank, and other accessories.

3.10.1 Starter. The engine shall be provided with a rewind-type starter with provisions for a rope-type starter, to be used in the event of failure of the rewind starter.

3.10.2 Exhaust muffler. The exhaust muffler shall be directed away from the compressor and engine air cooling intakes.

3.10.3 Crankshaft. The crankshaft speed shall not exceed 3,600 rpm.

3.10.4 Engine fuel. The engine fuel shall conform to MIL-G-5572, grade 80-145 or MIL-G-3056, types I and II. The specification number and grade shall be permanently marked on the fuel tank.

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3.10.5 Engine fuel tank. The engine fuel tank shall have a minimum capacity of 1-1/2 gallons.

3.11 Carrying handles. Two D-shaped tubular handles shall be provided for carrying the blower. Each handle shall be attached to one end of the base plate with two hinged fasteners so that the handles can be folded into either a vertical or horizontal position and locked. The handles shall be fastened to the base plate on the outside of the metal hood specified in 3.12. The height of the handles shall not exceed the overall height dimension specified in 3.16.

3.12 Metal hood. A metal hood shall be provided which can be used for preheating during low-temperature operations and for storage purposes when the blower is not in use. The hood shall fit inside the tiedowns and carrying handles to permit use of these provisions with the hood on. A 6-1/2-inch-diameter opening shall be strategically located so that the hot air from the heater will not impinge directly upon any components which might be adversely affected by high temperatures. An attached metal cover shall be provided to cover the opening when not in use. An opening shall be provided of such size that will permit four hoses each having a 1-3/8 inch outer diameter to pass through the hood and connect to the discharge header specified in 3.9. The opening shall be so constructed that the hood may be installed or removed without disconnecting any of the four hoses (the hood may be removed for hose connections). An attached metal cover shall be provided to cover the opening when the hoses are not connected. Clamps which will withstand the acceleration forces specified in 3.5.3.1 shall be provided for fastening the hood to the base plate.

3.13 Handbook and record compartment. A handbook compartment shall be provided as an integral part of the blower for containing the equipment operation handbook and maintenance and inspection record. The compartment shall be weatherproof if not contained in an inclosure and shall be a minimum size of 9 X 11-1/2 X 2 inches.

3.14 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

* 3.15 Radio-interference suppression. Radio interference shall not exceed the limits specified in MIL-STD-461.

3.16 Dimensions. The overall dimensions of the blower shall not exceed 25 inches in length, 20 inches in height, and 20 inches in width.

3.16.1 Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the blower, they shall be held or limited accordingly.

3.17 Weight. The weight of the blower, without gasoline and oil, shall not exceed 110 pounds.

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* 3.18 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.19 Finishes and protective coatings.

* 3.19.1 Magnesium alloy parts. Magnesium base alloy parts shall be treated with an anodic coating in accordance with MIL-M-3171, type IV, followed by two coats of zinc chromate primer in accordance with MIL-P-8585. Any surfaces not exposed to weather shall be given one coat of the primer and one coat of enamel conforming to MIL-E-7729, type I, and color 13538 of Bulletin 166. Surfaces exposed directly to the elements or to wear by handling shall be given two coats of enamel.

3.19.2 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.19.2.1 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.19.2.2 Where the primary purpose of the treatment is to afford a suitable paint base, chemical films conforming to MIL-C-5541 may be used in lieu of anodizing.

3.19.2.3 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film conforming to MIL-C-5541 in lieu of anodizing.

3.19.2.4 When abrasion resistance is a factor, chemical films conforming to MIL-C-5541 shall not be used in lieu of anodizing.

* 3.19.3 Inside surfaces. After the treatment specified in 3.19.2, aluminum alloy surfaces inside the hood shall be painted with one coat of primer conforming to MIL-P-8585 and one coat of enamel conforming to MIL-E-7729, type I, and color 13538 of Bulletin 166.

3.19.4 Exposed surfaces. Aluminum alloy surfaces directly exposed to the elements or to wear by handling shall be painted with the primer and two coats of the enamel specified in 3.19.3.

* 3.19.5 Ferrous parts. Bolts or parts that are subject to corrosion shall be cadmium plated in accordance with QQ-P-416, type I, 2, or 3, as applicable, and of a class that is adequate to achieve the degree of protection required.

* 3.19.5.1 Surfaces of ferrous parts to be painted shall be sandblasted or pickled to produce a clean surface, and painted with one coat of primer conforming to TT-P-664. Surfaces directly exposed to wear or elements shall be painted with two coats of enamel conforming to MIL-E-7729, type I, and color 13538 of Bulletin 166. Internal surfaces shall be given one coat of the enamel.

3.20 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.21 Marking. Brief instructions for starting, operating, and stopping the blower shall be permanently located in a readily visible position on the engine. Lubrication and servicing of main components shall be included in the instructions. The instruction plate shall be so located that oil, dirt, water, et cetera, will not affect its legibility nor cause it to become obliterated.

3.21.1 Filters. Fuel and intake-air instructions as to the method and period of cleaning shall be clearly marked on the filters.

3.21.2 Oil. Oil to be used shall be marked on the blower and shall conform to MIL-L-2104 or MIL-L-6082.

3.22 Workmanship.

3.22.1 General. The blower, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

a. Freedom from blemishes, defects, burrs, and sharp edges.

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.

c. Thoroughness of soldering, welding, brazing, painting, wiring, and riveting.

d. Alignment of parts and tightness of assembly screws and bolts; et cetera.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of blowers shall be classified as follows:

a. Quality conformance inspection . . . . . (See 4.3)

b. Preproduction tests . . . . . (See 4.6)

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* 4.3 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plan and test.

4.3.1 Individual tests. Each blower shall be subjected to the following tests as described under 4.5.

a. Examination of product

b. Functional check

4.3.2 Sampling plan and test.

4.3.2.1 Lot. A lot shall consist of blowers manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.3.2.2 Sampling plan. One blower shall be selected at random from each lot of 100 or fraction thereof produced and subjected to the functional check described under 4.5.2 except the test time shall be 75 hours.

4.3.2.3 Rejection and retest. When one blower selected from a lot fails to meet the specification, no blowers still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.4 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of blowers on hand or later produced shall not be made until it is determined that they meet all the requirements of the specification.

4.3.2.5 Defects in blowers already accepted. The investigation of a test failure could indicate that defects may exist in blowers already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and methods of correcting them.

4.4 Test conditions.

4.4.1 Instruments. All instruments used for test measurements shall be entirely suitable for use with the equipment being tested under the specified conditions.

4.4.2 Standard atmospheric conditions. Unless otherwise specified, tests shall be made at atmospheric pressure (approximately 29.92 inches mercury (Hg)) and at room temperature (approximately 77°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

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4.4.3 Airflow. Airflow shall be measured through smooth-approach or flat-plate orifices in accordance with ASME Power Test Code 19.5:4 (see 6.3). ASME Power Test Code 9-1954 shall be used as the guiding document for the airflow-list procedure.

4.4.4 Temperatures. Temperatures shall be measured by means of thermocouples and by properly calibrated potentiometers. Insofar as practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit.

4.4.5 Pressures. Insofar as practicable, pressures and pressure differentials shall be measured by means of liquid manometers.

4.4.6 Data recording. Chronological data recording shall be accomplished at a maximum of 30-minute intervals during the performance and endurance testing and at 15-minute intervals during other operational type testing. All performance type tests (including start and stop) shall consist of a minimum of five recordings.

4.4.6.1 The minimum data for maintaining a chronological record during all performance-type tests shall consist of the following:

a. Item, type of test, data sheet number, et cetera. This information shall be included in the heading.

b. Time - will reflect recording interval.

c. Engine instrumentation - air intake (°F), cylinderhead temperature (°F), oil sump temperature (°F, and engine rpm).

d. Blower component - oil sump temperature (°F), air intake temperature (°F), and manifold pressure (inches Hg).

e. Flow measuring duct - air temperature (°F) (position shall not disturb airflow through the orifice), static pressure (upstream) (inches H_2O), differential pressure across orifice (inches H_2O).

f. Ambient conditions (surrounding blower assembly including test cell conditions) - barometric pressure (inches Hg), dry-bulb temperature (°F), wet-bulb temperature (°F). The wet-bulb temperature may be omitted during low temperature testing.

NOTE: Ambient barometric pressure to be recorded at maximum intervals of 4 hours and test cell pressure at regular recording intervals.

g. Total time - for use during endurance testing to show running record of accumulated time. Endurance should not include time accumulated during other testing.

h. Remarks - shall include records of reference to other data, such as oil and fuel consumed, servicing, adjustments, malfunctions, corrective measures, photographs, et cetera.

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4.4.7 Running-in. The nature and extent of running-in shall be as recommended by the manufacturer, but shall not exceed 25 hours.

4.5 Test methods.

4.5.1 Examination of product. Each blower shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, lubrication, finish and marking.

4.5.2 Functional check. The blower shall be operated for a period of not less than 1 hour with the compressor operating against a pressure of 5 psig and shall have a free air delivery of not less than 50 cfm. The engine shall demonstrate adequate power and shall operate smoothly during normal loading and unloading of the blower. Complete instrumentation and a recording shall be as described under 4.4.

4.5.2.1 Relief valve. After completion of all functional checks, the relief valve shall be tested to determine compliance with 3.9.1.

4.5.3 Radio interference suppression. The blower shall be subjected to radio interference suppression tests in accordance with MIL-STD-461 except that during the radiated noise measurements, the measuring device shall be 6 feet from the blower.

* 4.5.4 Operation in tilted position. The blower shall be tilted in different positions at an angle of 15 percent (8-1/2°) with the horizontal plane. The blower shall be started and run in numerous positions and then allowed to stand for 4 hours in the tilted position most conducive to gasoline leakage from the carburetor into the cylinder. At the end of this time, the induction system shall be examined and the spark plug removed to permit examination of the cylinder for gasoline leakage. During the operational phase of this test, airflow instrumentation is not required. Engine and blower instrumentation and data recording shall be as covered under 4.4.

4.5.5 Environmental tests. The blower shall be subjected to the following environmental tests:

4.5.5.1 Cold start test. The blower shall be subjected to an ambient temperature of 0°F for a 24-hour period. At the end of this period, with temperature maintained, the blower shall be started without oil dilution or any external aid and then idled in accordance with the manufacturer's instructions until the proper operating temperature has been reached. While still at 0°F ambient temperature, the blower assembly shall be subjected to a functional check in accordance with 4.5.2.

* 4.5.5.2 Low temperature tests. The blower shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502.1; then the temperature will be raised to -65°F. After the temperature has stabilized at -65°F, the blower may be preheated, for a period not to exceed 30 minutes, by a Type H-2 heater conforming to MIL-H-4607. The heat shall then be removed. The blower shall start within 3 minutes and shall be subjected to a 2-hour (1 hour at

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minimum load and 1 hour at maximum rating) functional check in accordance with 4.5.2. The minimum load and full load operations shall be at governed speed. Five minutes after shut down, and without preheating, the blower shall be restricted within 3 minutes and then shut down after governed speed has been attained. At the completion of this test, and after the chamber temperature has been returned to normal ambient temperature, the blower shall be started and shall be subjected to a functional check in accordance with 4.5.2. Any distortion of parts, chipped or cracked paint, or other detrimental damage shall be cause for rejection.

* 4.5.5.3 High temperature tests. High temperature tests shall be in accordance with MIL-STD-810, Method 501.1, except that the blower shall not be operated. At the end of the 40-hour period, the temperature shall be lowered to 130°F. After the temperature has stabilized and while at this temperature, the blower shall be subjected to a functional check in accordance with 4.5.2. In addition to the functional check, the blower shall be inspected for distortion, chipped and cracked paint, or any other detrimental damage after the temperature has returned to normal ambient.

4.5.5.4 Altitude. The blower shall be subjected to a functional check in accordance with 4.5.2 at an altitude of 6,000 feet and an ambient temperature between 70° and 90°F.

* 4.5.5.5 Rain. The rain test shall be conducted in accordance with MIL-STD-810, Method 506.1. The equipment shall then be inspected for evidence of water in the fuel and oil tanks. The blower shall be cleaned, without disassembly, and subjected to the functional check specified in 4.5.2.

* 4.5.5.6 Sand and dust. The sand and dust test shall be in accordance with MIL-STD-810, Method 510.1. At the conclusion of the test, the blower shall be inspected for evidence of sand and dust in areas where it would be detrimental to blower operation. The blower shall then be subjected to the functional test specified in 4.5.2.

* 4.5.5.7 Salt spray. The salt spray test shall be in accordance with MIL-STD-810, Method 509.1. At the completion of the exposure, the blower shall be inspected for damage. The blower shall then be cleaned (rinsed and dried without disassembly) and subjected to the functional test specified in 4.5.2.

* 4.5.5.8 Fungus resistance. The fungus resistance test shall be conducted in accordance with procedure I of MIL-STD-810, Method 508.1. Following the test, the blower shall be inspected for damage. The blower shall then be subjected to the functional check specified in 4.5.2.

4.5.6 Acceleration loading. With the blower secured by the fittings specified in 3.5.2.1.1, it shall be subjected to an acceleration of 3G applied independently along each of the three major axes in either direction. Permanent deformation of structural members shall be cause for rejection.

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4.5.6.1 Emergency landing accelerations. The blower shall be subjected to the following accelerations in the order specified. The blower shall not break loose from the tiedowns and the attachment provisions shall not separate from the rest of the structure; however, the item need not be serviceable after being subjected to these accelerations. If it can be shown by computation that the equipment and all components conform to the requirements specified herein after sustaining the following accelerations, the computation may be substituted for the test.

a. 4-1/2G vertically downward that impose a load on the base in a downward direction.

b. 8G applied independently along each horizontal axis.

4.5.7 Endurance test. The endurance test shall consist of 300 hours of operation at governed rated speed. The testing shall be divided into two phases as follows:

a. Phase I - 280 hours

200 hours at 3-1/2 psig delivery pressure on the compressor
80 hours at 5 psig delivery pressure on the compressor

b. Phase II - 20 hours

5 hours at no-load
5 hours at 5 psig
5 hours at no-load
1 hour at no-load and 10 percent overspeed
1 hour at no-load
3 hours of alternate starting and stopping (5-minute intervals)

4.5.7.1 Readings of all pertinent data shall be taken during the test run. The oil system shall be drained and refilled with new oil at the start of the endurance test and as often thereafter as recommended by the manufacturer. Oil shall be added no oftener than every 25 hours. The ignition breaker points, spark plug, and filters may be inspected at the completion of each 50 hours of operation, but no adjustments shall be made unless necessary.

4.5.7.2 Stops. The blower shall be stopped at least every 25 hours and allowed to set for 15 minutes.

4.5.7.3 Test time. When terminated by failure, intervals of endurance test operation of less than 1 hour shall not be credited to the required test time.

4.5.8 Performance check. Upon completion of the endurance test, the blower shall be operated at a pressure of 5 psig. The actual free air delivery of the blower shall be not less than 95 percent of that specified in 4.5.2.

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4.5.9 Disassembly inspection. After the performance check specified in 4.5.8 has been completed, the blower shall be disassembled to the maximum practical extent to determine failures, distortion, excessive wear, weakness, or other unsatisfactory conditions. If measurements are made to determine wear, they shall be compared with similar measurements made prior to the tests, or with dimensions on the manufacturer's drawings. If unsatisfactory conditions exist which would be detrimental to continued operation or life of the compressor, the blower shall be rejected.

4.6 Preproduction testing.

4.6.1 Preproduction test sample tested by the contractor. The contractor shall subject one blower to the preproduction tests specified in 4.6.4.

* 4.6.2 Preproduction test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.6.2.1 The manufacturer shall furnish complete information regarding instructions and precautions necessary for starting and operating the blower at the temperatures specified in 4.5.5.1 through 4.5.5.3.

4.6.3 Preproduction test sample for the procuring activity. Along with the preproduction test report, the contractor shall submit a sample to the procuring activity which it will use as follows:

a. For a review of the mechanical construction of the product.

b. To perform any tests included in the specification, after reviewing the contractor's test report.

4.6.4 Preproduction tests. Preproduction tests shall consist of all tests described under 4.5.

* 4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A. The blowers shall be preserved and packaged in accordance with Method 11a of MIL-P-116. The unit container shall be as specified in 5.2.1.

* 5.1.2 Level C. The blowers shall be preserved and packaged to afford adequate protection against corrosion, deterioration and physical damage during shipment to the first receiving activity. This level may be the manufacturer's commercial practice if such practice conforms to the requirements specified above.

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* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

* 5.2.1 Level A. The blowers preserved and packaged as specified in 5.1.1 shall be packed in overseas type shipping containers conforming to PPP-B-601. The containers shall be of uniform size and shape and minimum tare and cube. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

* 5.2.2 Level B. Level B packing shall be the same as level A except that the shipping container shall be domestic type.

* 5.2.3 Level C. The blowers packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

* 5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186.

* 5.4 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type A-1 blower covered by this specification is intended for use in furnishing compressed air for the inflation of aircraft lifting bags.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing (see Section 5).

6.3 Airflow measurement. The airflow measuring device may be in accordance with Air Force Technical Report No. 4552, revision 1, in lieu of the device in accordance with ASME Power Test Code 19.5:4 (see 4.4.3).

* 6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 1730-F074

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. *This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.* Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426
1 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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